Explain the difference between Java Persistence API, Hibernate & Spring Data JPA

Java Persistence API (JPA)	Hibernate	Spring Data JPA
JPA is a Java specification that defines standards for mapping Java objects to relational databases.	Hibernate is an ORM framework that implements the JPA specification and provides additional features beyond the specification.	Spring Data JPA is a Spring module that simplifies working with JPA by reducing boilerplate code and providing ready-to-use repository support.
JPA standardizes persistence across different ORM implementations to ensure portability.	Hibernate provides the actual ORM functionality for mapping Java objects to relational database tables.	Spring Data JPA improves developer productivity by simplifying data access and repository creation.
JPA is only a specification and does not contain any executable implementation.	Hibernate is a complete implementation of JPA and can also be used independently without JPA.	Spring Data JPA is an abstraction layer built on top of JPA implementations such as Hibernate.
JPA cannot perform database operations by itself because it requires an implementation provider.	Hibernate performs database operations by implementing the JPA specification and generating SQL queries automatically.	Spring Data JPA delegates database operations to a JPA provider such as Hibernate.
JPA provides APIs such as EntityManager for performing CRUD operations	Hibernate provides CRUD operations through its implementation of EntityManager and Session	Spring Data JPA provides built-in CRUD methods through interfaces like JpaRepository, eliminating the need to write implementation code
JPA supports JPQL, Criteria API, and native SQL queries.	Hibernate supports JPQL and also provides Hibernate Query Language (HQL), Criteria API, and native SQL queries.	Spring Data JPA supports JPQL, native SQL, and automatically generates queries from repository method names.